2 **(a)** State what is meant by an *ideal gas*.

....

....

....

.... [2]

(b) A piston contains monoatomic ideal gas. The properties of the gas are shown in Table 2.1.

pressure / Pa	2.12×10^{7}
volume / m^3	1.84×10^{-2}
mass / kg	3.20
temperature / °C	22.0

Table 2.1

(i) Determine the root-mean-square speed of the gas atoms.

speed = ... $m\ s^{-1}$ [2]

For Examiner's Use

(ii) The volume of the gas in the piston increases suddenly. Using the first law of thermodynamics, explain what happens to the kinetic energy of the gas.

....

....

....

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....

....

.... [4]

(iii) On Fig. 2.1, sketch the graph representing

1. the process in **(b)(ii)**. Label the graph **A**

2. the process in **(b)(ii)** if the increase in volume is allowed to take place slowly. Label the graph **B**

Indicate the direction of the processes clearly.

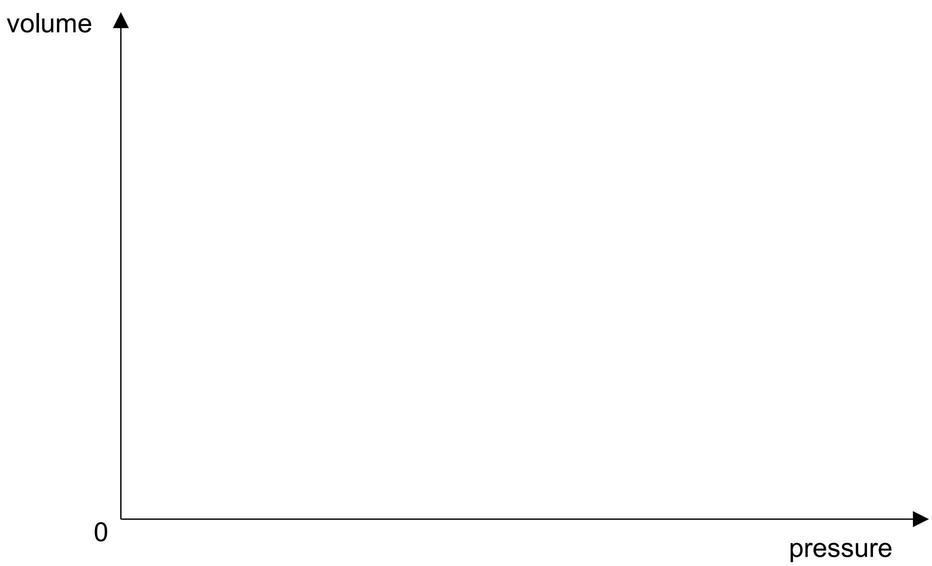

Fig. 2.1

[3]